Data Mining Project: Analyzing Reader Patterns and Genre Associations in the Goodreads Dataset*

* CS 4412: Data Mining

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***Abstract*—This proposal outlines a data mining project focused on the 2017 Goodreads dataset. This study aims to discover relationships between book genres, user behaviors, and how readers structure their reviews. We will be applying different techniques such as Association Rule Mining, K-Means Clustering, and Latent Dirichlet Allocation. This proposal seeks to understand how readers associate different genres in an attempt to define different segments of the reading community.**

***Index Terms*—Data Mining, Goodreads, Association Rules, Clustering, Text Mining, Pattern Discovery**

I. Introduction

The digital age has transformed reading from a personal, solo activity into a social activity. Various platforms like Goodreads and Fable provide a massive database of user-generated data through ratings, reviews, and custom shelves. While many similar projects exist to predict user's rating there is more value in understanding how users engage with literature. This project focuses on the UCSD Goodreads dataset to find these patterns.

II. Description

This project will utilize the Goodreads Book Datasets (2017) curated by Julian McAuley at UCSD, Rishabh Misra and Ndapap Nakashole

A. A. Source and Scale

- **Source:** https://cseweb.ucsd.edu/~jmcauley/datasets/goodreads.html
- **Size:** The full dataset contains metadata for 2.36 million books and 15.7 million detailed review texts. For this project, a subset (e.g., the "Young Adult" or "History" category) will be utilized to maintain computational feasibility while ensuring a sample size of over 100,000 records.
- **Format:** Data is provided in the CSV and json format, requiring significant preprocessing and flattening.

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B. Key Features and Schema

The dataset is split into two primary components:

1) **Metadata (*goodreads_books.json*):** Contains `book_id`, `title`, `authors`, `average_rating`, and importantly, `popular_shelves` (user-defined tags).
2) **Reviews (*goodreads_reviews.json*):** Contains `user_id`, `book_id`, `rating`, `review_text`, and `n_votes`.
3) **User Interactions (*goodreads_interactions.csv*):** Contains `user_id`, `book_id`, `is_read`, `rating`, and `is_reviewed`.

III. C. Data Quality and Preprocessing

Known issues include many user-defined shelves that share common tags like "to-read" and varying review lengths. This is most likely caused from Goodread's ability to easily add books to user's "to-read" shelve. Preprocessing will involve tokenization, stop-word removal for text mining and transforming the nested JSON shelves into a transaction-style for association mining.

IV. Discovery Questions

This project moves beyond prediction models to investigate the following questions the team had:

- What are the hidden associations between specfic book genres and user-defined "shelves"? (Do readers of "Romantasy" (Romance Fantasy) also show high interset in other genres like fantasy?
- Can we identify distinct Reader Personas based on interaction metrics? By clustering users based on their average rating variance, review frequency, and review length our goal is to discover if it is possible to classify readers as a critical reviewer or casual consumer.
- What type of patterns emerge from highly-voted reviews compared to low-voted ones?

V. Planned Techniques

The analysis will follow a multistage data mining pipeline as visualized in Figure 1.

Analysis Pipeline Diagram

1. Data Ingestion: Books & Reviews JSON Parsing

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2. Preprocessing: Admin Tag Cleaning & Language Filtering

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3. Mining Modules & Evaluation:

→ *Shelf Tags* ⇒ **FP-Growth** ⇒ Genre Associations (DQ1)
→ *User Metrics* ⇒ **K-Means** ⇒ Reader Personas (DQ2)
→ *Review Text* ⇒ **LDA** ⇒ Helpfulness Patterns (DQ3)

Fig. 1. Planned Data Mining Workflow mapping specific data features to their respective algorithms and discovery questions.

A. Association Rule Mining (FP-Growth)

We will treat each user's shelves as a transaction. Using the FP-Growth Algorithm to discover rules with high lift and confidence to identify non-obvious genre relationships. Our goal is to move beyond the standard "Fantasy" label and discover more sub-genre patterns.

B. Clustering (K-Means)

To answer the second discovery question, we will apply K-Means clustering on standarized numerical features derived the user dataset. We will be using the "Elbow Method" and "Silhoutette Score" to determine the optimal number of reader segments.

C. Text Mining (LDA - Topic Modeling)

Using the Latent Dirichlet Allocation (LDA) algorithm, we will perform topic modeling on the `review_text`. This will allow us to discover the primary topics of discussion within different genres without pre-defining what those topics are.

VI. Preliminary Timeline

The project will proceed according to the following milestones

TABLE I
PROJECT MILESTONE TIMELINE

Milestone	Period	Key Activities
M2	Weeks 8	Applying KDD process and Association Rule Mining
M3	Weeks 11	Clustering Analyses, Classification Techniques, detecting Anomalies, and Communicating results
M4	Weeks 14	Text Mining (LDA), Final Report

A. Anticipated Challenges

- Our primary challenge is lack of experience in analyzing data sets.
- Another challenge is the volume of data. The Goodreads dataset is comprised of multiple csv and jsons files that is multiple gigabytes compressed. Handling this in local memory will require using pandas chunking or focusing on smaller genre subsets. Text data can also be "noisy" and will require filtering to produce meaningful topics.
- Due to Goodreads' global reach our team expects to encounter non-English reviews, books, and shelf titles. Our team will have to strip any non-English text out of the datasets in order to prevent any of our proposed tests from failing.

VII. Milestone 2

This section details the work done for Milestone 2 of this data mining project. This milestone focused on an inital implementation of an exploratory discovery analysis to initally answer one of our three discovery questions. The section begins by explaining the reasoning for preprocessing and cleaning up the noisy dataset. Our team then analyses the most frequent tags of users' bookshelves to gain a first look at how users cateogrize books on the site. This section also covers our findings on discovery question 1 and how we came to our conclusions. In addition to discovery question 1, we have provided visualzaitons of our findings.

A. Preprocessing and Data Transformation

In order to transition from the introductory and theoretical discovery questions from the previous sections our team implemented multiple scripts to select and gather the raw data from the dataset. To begin answersing our discovery questions we will need to preprocess our data through a python code to filter the massive dataset by cleaning some administrative tags and filtering out non-English languages. First we performed a keyword search across the "popular_shelves" attribute of the `goodreads_books.json.gz` file. We specifically search for genres: 'romantasy', 'romantic-fantasy', 'fantasy-romance', 'romance', 'fantasy', and 'romantic'. The filtered results are stored in `01_filteredbooks.json`. The resulting json consists of records: book_id, title, average_rating, shelves_list.

To answer discovery question 1, we need to examine the different user interactions, so we implemented a second filtering script utilizing the book_id from our previous script to extract reviews from the `goodreads_reviews_dedup.json.gz` dataset, storing the output in `03_extractedreviews.csv`.

The datasets required significant cleaning before our team could conduct meaningful mining. First, we removed generic administrative tags like `'to-read'` and `'ebooks'` from the books dataset to reduce noise. Second, to focus our analysis and ensure data comprehension, we removed any non-English records from the reviews dataset using the Python package `langid`.

B. Exploratory Data Analysis

Before applying our mining algorithms, our team examined the distributions of ratings, community-defined tags (for example users can create virtual bookshelves and give them

a name), and review engagement to establish a baseline understanding of the "Romantasy" and other romance genres subset.

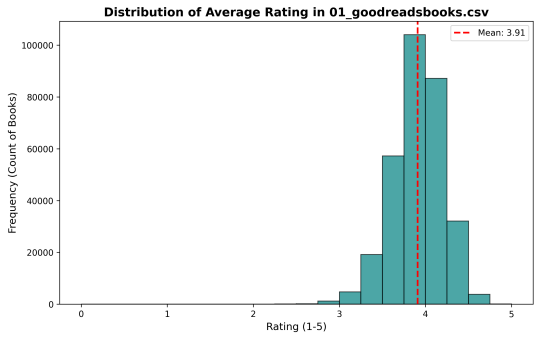

Fig. 2. Distribution of books rating in 01_goodreadsbooks.csv

Examining the histogram of the average ratings of romance books in the 01_goodreadsbooks.csv we find that most readers rated the books around 4. This suggests that romance readers find books great reads. They often are not critical of books.

1) Shelf Tag Distribution: Analysis of the shelves_list indicates that the tag 'to-read' is the most frequent attribute. During the preprocessing phase, we implemented a filter to only show tags that correspond to actually refer to book descriptions and not generic tags such as "to-read". However after during the postprocessing phase the most prevalent tags included fiction, romance, and fantasy. This aligns with the "Romatansy" genre which confirms Romantasy is the most pollinated genre.

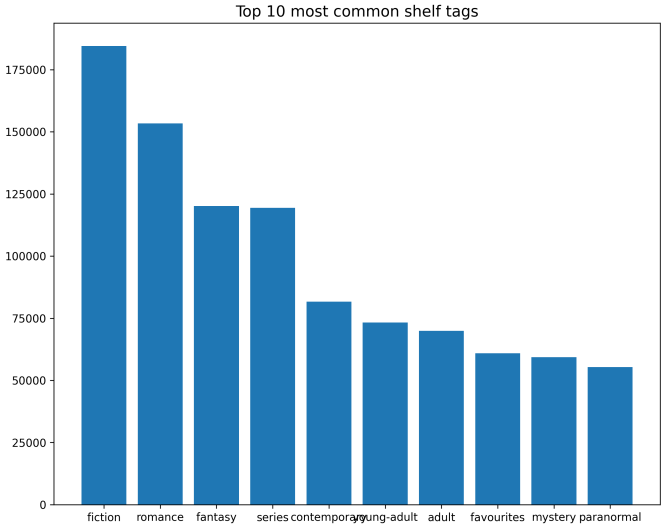

Fig. 3. Distribution of the top 10 most frequent shelf tags post-cleaning

2) Engagement Volume: Beyond genre tags and ratings, we extracted user engagement metrics from the `goodreads_reviews_dedup.json.gz` dataset to identify behavioral patterns. A Pearson correlation analysis reveals a notable moderate positive correlation of 0.52 between the number of comments (`n_comments`) and the number of votes (`n_votes`), suggesting that reviews sparking discussion are also more likely to be perceived as helpful or popular. Conversely, our initial hypothesis that `review_length` would drive higher engagement was unsupported, as it maintained negligible correlations with both `n_votes` (0.03) and `n_comments` (0.04). Furthermore, the numerical `rating` appears entirely decoupled from interaction volume, showing correlations of 0.00 and -0.01 against votes and comments, respectively. These results indicate that while social interactions (voting and commenting) tend to scale together, they are driven by factors other than the sentiment of the rating or the verbosity of the review.

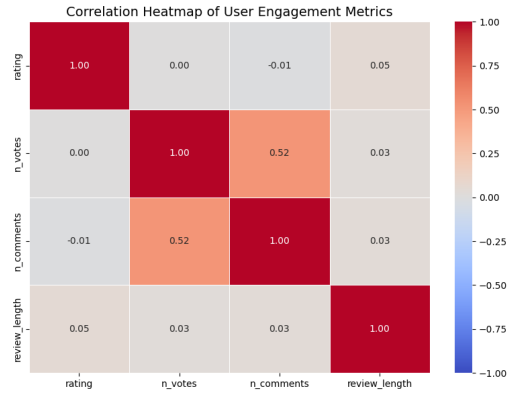

Fig. 4. Correlation Heatmap of User Engagement Metrics

Visualizing this relationship via a scatterplot (Figure **??**) confirms the heatmap findings. While the majority of reviews cluster near the origin of the graph, there is a clear positive trend where reviews exceeding $1,000$ votes frequently receive over 150 comments. These results indicate that while social interactions (voting and commenting) tend to scale together, they are driven by community relevance of a review rather than the sentiment of the rating or the content of a certain book.

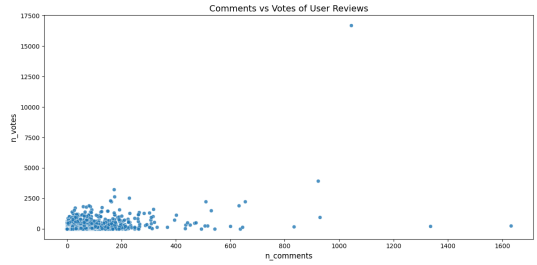

Fig. 5. Scatterplot of Comments vs. Votes

C. Answering Discovery Question 1: Genre Associations

Our team originally asked if readers of "Romantasy" (Romance Fantasy) also showed high interest in other specific romance sub-genres. After running the FP-Growth algorithm on our transaction data, the top association rules generated lift

values of approximately 6.8 to 7.0 for various fantasy sub-genres, rather than romance genres.

TABLE II
TOP DISCOVERED GENRE ASSOCIATION RULES BY LIFT

Antecedents	Consequents	Lift	Support
{paranormal, fiction}	{supernatural}	7.59	0.088
{favourites, novels}	{novel}	6.93	0.081
{novels, adult}	{adult-fiction}	6.44	0.082
{paranormal, fantasy}	{supernatural}	6.23	0.093

While we did not see strong associations between romance genres at our current minimum support threshold of 0.08, our findings strongly support the broader hypothesis of sub-genre cross-pollination. As shown in Table **??** readers of fantasy and fiction are highly tag other related genres. This is evidenced by the strong associations found between "paranormal" and "supernatural" fiction yielding a high lift of 7.59. Another pattern we observed is the overlap between "paranormal" and fantasy yielding a high lift of 6.23 which confirms that sub-genre crossover pollination is a real thing.

Fig. 6. Genre Associations

VIII. UPDATED PLAN FOR MILESTONE 3

Moving into Milestone 3, our plan remains aligned with the original Milestone 1 proposal. Since we have successfully established our preprocessing pipeline and validated our association rules, our next focus will be applying K-Means clustering to answer our second discovery question. We will collect user interactions from rating variance and review length—to identify distinct reader categories (e.g., casual consumers vs. critical reviewers). Additionally, we will begin exploring Latent Dirichlet Allocation (LDA) for text mining the review bodies.

ACKNOWLEDGMENT

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